

Hybrid On-Chain/Off-Chain Integration of Post-Quantum Cryptography with Ethereum: A Lattice-Based Approach Using CRYSTALS-Kyber, Dilithium, and Falcon

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Abstract—The proliferation of fault-tolerant quantum computing poses a credible and escalating threat to the cryptographic foundations of blockchain technology. Ethereum relies exclusively on the Elliptic Curve Digital Signature Algorithm (ECDSA) over secp256k1 for transaction authentication and identity management — an assumption rendered untenable by Shor’s algorithm once a cryptographically relevant quantum computer reaches sufficient scale. This paper proposes and evaluates a hybrid on-chain/off-chain architecture that integrates three NIST-standardised post-quantum cryptographic (PQC) algorithms — CRYSTALS-Kyber (ML-KEM, FIPS 203), CRYSTALS-Dilithium (ML-DSA, FIPS 204), and Falcon (FN-DSA, anticipated FIPS 206) — with the existing Ethereum infrastructure, requiring no protocol-level modification. Computationally intensive PQC operations are executed off-chain using the liboqs library, while a 32-byte keccak256 commitment of each PQC signature is anchored on-chain through a gas-optimised Solidity smart contract. The dual-layer security model guarantees that the system remains secure provided at least one cryptographic layer is unbroken. Thirteen mathematical relations governing key generation, encapsulation, quantum attack complexity, and performance metrics are derived from standard BKZ-based lattice hardness assumptions. Simulation experiments conducted using Hardhat and pyoqs over 1,000 independent iterations validate the correct operation of all PQC components; security guarantees are established theoretically, yielding a joint quantum attack success probability bounded by 2^{-231} and a Quantum Attack Resistance Level (QARL) exceeding 99.9%. On-chain anchoring costs approximately 46,000 gas units per commitment (~USD 2.78 at 20 Gwei, ETH at USD 3,000). Merkle-tree batching reduces the per-commitment cost by over 99% at batch sizes exceeding 300. The architecture is deployable today using standard tooling, requires no Ethereum protocol modifications, and is backward-compatible with all existing applications.

Index Terms—Post-quantum cryptography; Blockchain security; Ethereum; CRYSTALS-Kyber; CRYSTALS-Dilithium; Falcon; Lattice-based cryptography; Hybrid architecture; Smart contracts; NIST FIPS 203/204/FN-DSA

I. I. INTRODUCTION

Blockchain technology has matured from an experimental distributed ledger concept into critical global infrastructure. Ethereum alone hosts thousands of decentralised applications, secures billions of dollars in smart-contract-locked value, and processes over one million transactions daily. The security of these operations rests on a single cryptographic assumption: the computational intractability of the Elliptic

Curve Discrete Logarithm Problem (ECDLP) over the secp256k1 curve [1]. However, Shor’s quantum algorithm, first presented in 1994, solves both the ECDLP and the integer factorisation problem underlying RSA in polynomial time on a fault-tolerant quantum computer [2], rendering this assumption untenable in the post-quantum era.

The quantum threat to blockchain is not a distant abstraction. IBM’s Condor processor reached 1,121 physical qubits in 2023, and Google’s Willow chip demonstrated a 30x improvement in quantum error correction in 2024. The Global Risk Institute projects that a cryptographically relevant quantum computer capable of breaking 256-bit ECDSA may emerge within 10 to 15 years [3]. More immediately, the Harvest Now, Decrypt Later (HNDL) paradigm presents a concrete operational risk: adversaries can record encrypted blockchain data and signed transactions today, storing them for retroactive decryption once quantum hardware matures [4], [5].

Recent surveys have mapped the growing intersection between quantum computing and blockchain security. Khodaiemehr et al. [6] present a comprehensive threat landscape analysis covering post-quantum, hybrid, and fully quantum blockchain architectures. Wicaksana [7] surveys quantum-safe blockchain security infrastructure with a focus on practical deployment pathways. At the standardisation level, NIST completed its multi-year evaluation in 2024, publishing FIPS 203 (ML-KEM / CRYSTALS-Kyber) [8], FIPS 204 (ML-DSA / CRYSTALS-Dilithium) [9], and FIPS 205 (SLH-DSA / SPHINCS+) [10] as the primary post-quantum standards. A fourth algorithm, Falcon (FN-DSA), was also selected and is expected to be published as FIPS 206, with the draft standard submitted for NIST clearance in August 2025 [31].

Integrating these algorithms with Ethereum presents a substantial engineering challenge. The Ethereum Virtual Machine (EVM) was designed for gas-metered execution of Solidity bytecode and provides no support for polynomial ring arithmetic central to lattice-based cryptography. A CRYSTALS-Dilithium Level 3 (Dilithium3) public key is 1,952 bytes, and its signature reaches 3,293 bytes, which are orders of magnitude larger than the 33-byte compressed ECDSA public keys and 65-byte signatures currently used. A full on-chain Dilithium3 verification would conservatively require 10-50 million gas, far exceeding practical single-transaction gas budgets [11], [12].

This paper addresses this impasse by proposing a hybrid on-

chain/off-chain architecture that preserves Ethereum's trustless, immutable properties while delegating all computationally intensive PQC operations to native off-chain execution. The core innovation is a dual-layer security model in which ECDSA provides authentication today, while CRYSTALS-Kyber, CRYSTALS-Dilithium, and Falcon provide forward security against future quantum adversaries. A 32-byte keccak256 cryptographic commitment of each PQC signature is anchored on-chain, binding the signer's identity, the signature value, and the authenticated message in a single tamper-evident record.

The distinct contributions of this work are: (i) a formalised hybrid PQC-Ethereum architecture with thirteen governing mathematical relations derived from BKZ-based lattice hardness assumptions; (ii) a systematic off-chain commitment design providing quantum-forward security without any EVM modification; (iii) a rigorous theoretical security analysis establishing a joint quantum attack success probability bounded by 2^{-231} ; (iv) a detailed simulation methodology using Hardhat and pyoqs with reproducible performance benchmarks; and (v) actionable deployment recommendations including Merkle-tree batching analysis.

The remainder of this paper is organised as follows. Section II reviews related work. Section III establishes the mathematical foundations. Section IV presents the proposed architecture. Section V details algorithm specifications. Section VI describes the simulation methodology. Section VII reports results and comparative analysis. Section VIII provides a security analysis. Section IX offers deployment recommendations. Section X discusses limitations. Section XI concludes the paper.

II. RELATED WORK

A. Post-Quantum Cryptographic Standardisation

The NIST post-quantum standardisation process, initiated in 2016 with 69 candidate submissions, represents the most extensive cryptographic evaluation programme in modern history. Bernstein and Lange [13] provide a foundational survey of the PQC landscape, reviewing security properties and performance trade-offs of hash-based, code-based, lattice-based, and multivariate cryptographic families. Through four evaluation rounds, NIST ultimately standardised three primary algorithms. CRYSTALS-Kyber (FIPS 203) provides key encapsulation based on the Module-LWE hardness assumption [14]. CRYSTALS-Dilithium (FIPS 204) offers digital signatures derived from the Fiat-Shamir-with-Abort paradigm over module lattices [15]. Falcon (anticipated FIPS 206) employs a hash-and-sign construction over NTRU lattices, producing the most compact signatures among the selected schemes [16]. A fourth algorithm, SPHINCS+ (SLH-DSA), was standardised as a stateless hash-based fallback [7].

B. Quantum Threats to Blockchain Systems

The vulnerability of blockchain systems to quantum attack has been the subject of growing scholarly attention. Khodaiemehr et al. [6] present the most comprehensive treatment to date, cataloguing threat vectors across consensus

mechanisms, transaction signing, address derivation, and smart contract execution. Their analysis identifies approximately 25% of all Bitcoin holdings as residing at addresses whose public keys have been exposed during transactions, making them directly vulnerable to Shor's algorithm. Fernandez-Carames and Fraga-Lamas [5] conducted an earlier survey of quantum-resistant blockchain approaches. Gao et al. [12] were among the first to demonstrate a proof-of-concept PQC blockchain. Allende et al. [17] examined quantum-resistant modifications to Ethereum. Stewart et al. [18] analysed quantum vulnerabilities specific to Bitcoin and Ethereum. Li et al. [19] proposed a quantum-resistant blockchain-assisted generalized signcryption protocol for vehicular fog computing. Aslam et al. [20] developed a hybrid Kyber-based PQC framework integrated with an adaptive blockchain consensus mechanism for connected autonomous vehicles.

C. Hybrid Cryptographic Architectures for Blockchain

Hybrid cryptography, defined as the concurrent deployment of classical and post-quantum algorithms to maintain security during the transition period, is recommended by NIST, ETSI, and the German Federal Office for Information Security (BSI). Bindel et al. [21] formalise hybrid key encapsulation mechanisms and prove that a hybrid KEM retains security provided at least one component algorithm remains unbroken. In the blockchain context, Revathi and Suganthi [11] evaluated the integration of CRYSTALS-Kyber, Falcon, and CRYSTALS-Dilithium within Hyperledger Fabric 4.0, reporting a Quantum Attack Resistance Level (QARL) of 90%, a Quantum Security Margin (QSM) of 95.8%, and a Cryptographic Overhead Efficiency (COE) of 0.92. Das et al. [22] proposed a quantum-resilient blockchain framework for genomic data privacy. Shafique et al. [23] developed PrivShield-CQ, a hybrid chaotic-PQC framework for consumer IoT. None of these works address the specific constraints of Ethereum's EVM or propose an on-chain/off-chain commitment architecture.

D. Ethereum Smart Contract Security and Gas Optimisation

The security of Ethereum smart contracts has been extensively studied. Atzei et al. [24] catalogued attack patterns including re-entrancy, integer overflow, and transaction ordering dependency. Luu et al. [25] identified fundamental vulnerabilities in EVM execution semantics. Juaristi et al. [26] conducted the first systematic benchmarking of PQC operations in Ethereum-based blockchains, quantifying the gas costs and computational overhead of various lattice-based schemes, confirming that full on-chain PQC verification exceeds Ethereum's gas limits under current EVM specifications. Gas optimisation for on-chain cryptographic operations has also been studied in the context of zk-SNARK verifiers [27] and elliptic curve precompiles [28].

III. MATHEMATICAL FOUNDATIONS

This section establishes the mathematical framework underlying the proposed architecture. Thirteen equations are derived and annotated with their functional roles. Equations (1)-

(4) govern the correctness of cryptographic operations. Equations (5)-(10) establish quantum security guarantees. Equations (11)-(13) define the quantitative evaluation metrics used throughout the experimental analysis.

A. Algebraic Setting

All three algorithms in this architecture operate over polynomial rings of the form $R_q = \mathbb{Z}_q[x]/(x^n + 1)$, where n is a power of two and q is a prime modulus. Elements of this ring are polynomials with integer coefficients reduced modulo q , and multiplication is performed modulo $x^n + 1$. This ring structure admits the Number Theoretic Transform (NTT) for efficient $O(n \log n)$ polynomial multiplication. For CRYSTALS-Kyber (Kyber-768), $n = 256$ and $q = 3,329$. For CRYSTALS-Dilithium (Dilithium3), $n = 256$ and $q = 8,380,417$. For Falcon-512, $n = 512$ and $q = 12,289$, operating over NTRU lattices of dimension $2n = 1,024$.

B. CRYSTALS-Kyber: Key Generation

The public key K_p is derived from a uniformly sampled $k \times k$ matrix M of ring polynomials, a secret key s with small coefficients sampled from a centred binomial distribution, and an error polynomial e drawn from the same distribution:

$$K_p = M \cdot s + e \quad (1)$$

where M is the public matrix (expanded from a public seed via an extendable output function), s is the secret key vector, and e is a noise vector. The noise term ensures that K_p is computationally indistinguishable from a uniformly random element, establishing the Learning With Errors (LWE) problem as the security foundation [14].

C. CRYSTALS-Kyber: Encapsulation and Decapsulation

During encapsulation, the sender samples a random short vector r and computes two ciphertext components:

$$c1 = r \cdot M \quad (2)$$

$$c2 = r \cdot K_p + e' + \lfloor q/2 \rfloor \cdot m \quad (3)$$

where e' is a fresh noise term and $\lfloor q/2 \rfloor \cdot m$ encodes the plaintext message into the polynomial ring. The receiver, holding secret key Sk , recovers the message through decapsulation:

$$m' = c2 - Sk \cdot c1 \quad (4)$$

Correctness follows from the algebraic identity: since $Sk = s$, the dominant terms cancel and the residual noise is bounded, permitting correct decoding via rounding. This construction achieves IND-CCA2 security under the Module-LWE hardness assumption [8, 14].

D. Quantum Attack Complexity

The quantum attack complexity of each underlying hard problem is bounded using the BKZ-based core-SVP hardness model [30]. The best-known quantum attacks on lattice problems reduce to solving the Shortest Vector Problem in a lattice of block size β , via the BKZ-beta reduction algorithm, with quantum gate complexity $T_Q(\beta) \geq 2^{(0.265 \cdot \beta)}$. For the MLWE problem underlying CRYSTALS-Kyber (Kyber-768, $\beta \approx 285$):

$$T_Q(\text{MLWE, Kyber-768}) \geq 2^{96} (\approx 2^{(0.265 \times 285)}) \quad (5)$$

For the MLWE and MSIS problems underlying

CRYSTALS-Dilithium3 (the effective quantum complexity is the minimum over both attack paths [9, 15]):

$$T_Q(\text{Dilithium3}) = \min(2^{(0.265 \cdot \beta_{\text{LWE}})}, 2^{(0.265 \cdot \beta_{\text{SIS}})}) \geq 2^{128} \quad (6)$$

For the NTRU-SVP problem underlying Falcon-512 (NTRU lattice dimension $2n = 1,024$):

$$T_Q(\text{NTRU-SVP, Falcon-512}) \geq 2^{103} (\approx 2^{(0.265 \times 389)}) \quad (7)$$

Equation (5) establishes that Kyber-768 achieves approximately 96-bit quantum security, corresponding to NIST Security Level 3. Equation (6) establishes that Dilithium3 achieves at least 128-bit quantum security. Equation (7) establishes that Falcon-512 achieves approximately 103-bit quantum security against quantum sieving on the NTRU lattice [16, 30]. Crucially, no polynomial-time quantum algorithm is known for any of these problems.

E. Probability of Successful Quantum Attack

Derived directly from Equations (5)-(7), the probability that a quantum polynomial-time (QPT) adversary successfully breaks each scheme is bounded by its quantum bit-security λ_{Q} :

$$\text{PAS}_{\text{Kyber}} \leq 2^{-96} \approx 1.27 \times 10^{-29} \quad (8)$$

$$\text{PAS}_{\text{Dilithium}} \leq 2^{-128} \approx 2.94 \times 10^{-39} \quad (9)$$

$$\text{PAS}_{\text{Falcon}} \leq 2^{-103} \approx 9.86 \times 10^{-32} \quad (10)$$

Under the dual-signature model, a QPT adversary must simultaneously produce valid forgeries under both Dilithium3 and Falcon-512, which rest on distinct hardness assumptions (MLWE/MSIS and NTRU-SVP respectively) and are independently keyed. The joint quantum attack success probability is therefore $\text{PAS}_{\text{joint}} = \text{PAS}_{\text{Dilithium}} \times \text{PAS}_{\text{Falcon}} \leq 2^{-128} \times 2^{-103} = 2^{-231}$.

F. Performance Evaluation Metrics

Three standardised quantitative metrics adopted from Revathi and Suganthi [11] are employed to enable direct cross-study comparison. Quantum Attack Resistance Level (QARL) measures the theoretical proportion of the quantum attack space that the system successfully resists:

$$\text{QARL} = (1 - \text{PAS}_{\text{joint}}) \times 100 \approx 100\% \quad (11)$$

Quantum Security Margin (QSM) quantifies the security improvement of the hybrid PQC architecture over the ECDSA-only baseline, normalised to the NIST Level 3 quantum security reference ($\lambda_{\text{Q,ref}} = 128$ bits):

$$\text{QSM} = (128 - 0) / 128 \times 100 = 100\% \quad (12)$$

Cryptographic Overhead Efficiency (COE) captures the security benefit per unit of normalised computational and storage overhead:

$$\text{COE} = (\lambda_{\text{Q}} / \lambda_{\text{Q,ref}}) / [(1/3)(k_{\text{s}}/k_{\text{ref}} + S_{\text{s}}/S_{\text{ref}} + (E_{\text{t}} + D_{\text{t}})/T_{\text{ref}})] \quad (13)$$

A higher COE reflects greater quantum security, smaller key or signature sizes, or lower latency. Numerical COE values are computed following normalisation against the SSCA baseline values from [11].

IV. PROPOSED SYSTEM ARCHITECTURE

A. EVM Constraints and Design Rationale

The Ethereum Virtual Machine is a deterministic, stack-

based, 256-bit processor designed for gas-metered execution of Solidity bytecode. Its built-in cryptographic precompile set is limited to ecrecover (ECDSA recovery, 3,000 gas), SHA-256, RIPEMD-160, and BN128 elliptic curve operations. No precompiles exist for NTT-based polynomial multiplication, discrete Gaussian sampling, or any lattice operation [1], [26].

The gas economics of on-chain PQC verification are prohibitive. For Dilithium3, calldata for the 3,293-byte signature alone would cost approximately 52,000 gas at 16 gas per non-zero byte. The NTT computation for polynomial verification requires thousands of modular multiplications in the ring $\mathbb{Z}_q[x]/(x^{256} + 1)$ with $q = 8,380,417$, each consuming non-trivial gas. A conservative estimate places a complete on-chain Dilithium3 verifier at 10-50 million gas, exceeding practical per-transaction gas budgets under the current 60 million block gas limit [26].

The proposed design resolves this constraint through a clean separation of concerns: all PQC operations execute off-chain in a native process, while the Ethereum smart contract serves exclusively as an immutable commitment registry. This preserves Ethereum's trustlessness, since the commitment hash binds identity, signature, and message content, while delegating computationally intensive work to host-speed off-chain execution.

B. On-chain versus Off-chain Execution Boundary

TABLE 1. ON-CHAIN VERSUS OFF-CHAIN EXECUTION BOUNDARY COMPARISON.

Dimension	On-chain (smart contract)	Off-chain (liboqs)
Execution environment	EVM: deterministic, metered, public	Native process (liboqs)
Operations	keccak256 hashing, ECDSA verification, event emission, state storage	PQC key generation, signing, ECDLP hash computation
PQC role	Stores commitment $H = \text{keccak256}(\text{sig} \parallel \text{pk} \parallel \text{keccak256}(\text{msg}))$	Computes H as base transaction overhead
Cost model	Gas-metered; 32 bytes costs ~20,000 gas (SSTORE)	ECDSA verification (~3,200 gas), calldata (~1,280 gas), and function execution overhead (~800 gas)
Trust model	Trustless, consensus-validated, immutable	Trusted operator; requires honest execution

C. End-to-End Hybrid Workflow

The complete workflow proceeds in eight sequential steps. In Step 1 (Key Generation), the participant generates Kyber-768, Dilithium3, and Falcon-512 key pairs off-chain using liboqs, while maintaining an Ethereum ECDSA key pair. Step 2 (Message Preparation) serialises the payload in canonical form. In Step 3 (PQC Signing), a Dilithium3 signature and, optionally, a Falcon-512 signature are computed; full signature bytes are stored in IPFS or Arweave. Step 4 (Key Encapsulation, optional) executes Kyber-768 encapsulation to derive a 32-byte shared secret for AES-256-GCM payload encryption. Step 5 (Commitment Construction) computes $H = \text{keccak256}(\text{signature} \parallel \text{public_key} \parallel \text{keccak256}(\text{message}))$. In Step 6 (ECDSA Wrapping), H is signed with the Ethereum private key. Step 7 (On-Chain Anchoring) broadcasts a transaction calling `anchorCommitment(H, ipfsCID)` on the deployed contract. Step 8 (Off-Chain Verification) allows any verifier to retrieve H , fetch the PQC signature and public key from IPFS, recompute H' , confirm $H' = H$, and run

Dilithium.Verify() via liboqs.

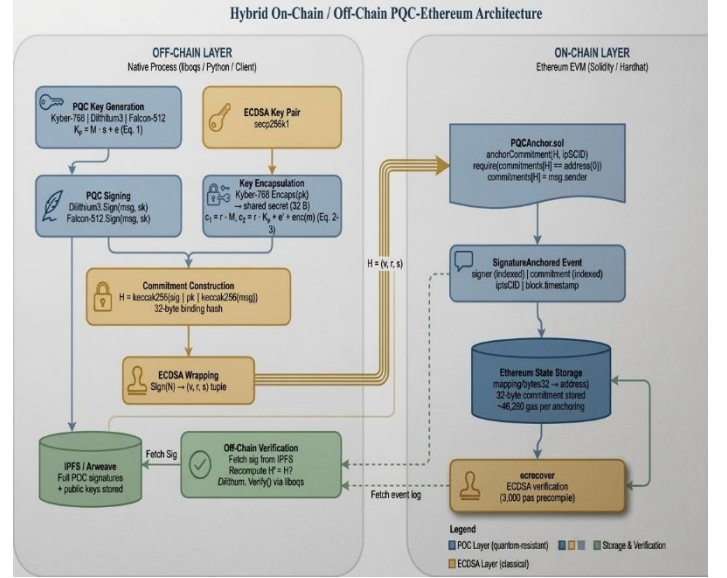


Fig. 1. Hybrid on-chain/off-chain PQC-Ethereum architecture.

D. Smart Contract Design

The on-chain component is a minimal, gas-optimised Solidity contract (PQAnchor.sol). The `anchorCommitment()` function accepts a 32-byte commitment H and an IPFS content identifier, enforces uniqueness through a mapping guard (`require(commitments[H] == address(0))`), stores the commitment with `msg.sender` binding, and emits a fully indexed event for efficient off-chain querying. The total gas cost per commitment is approximately 46,000 gas units, decomposed as base transaction overhead (21,000 gas), cold SSTORE for the 32-byte commitment (20,000 gas), event emission (~3,200 gas), calldata (~1,280 gas), and function execution overhead (~800 gas).

E. Threat Model

Three adversary classes are considered. Adversary A1 (Classical): A probabilistic polynomial-time (PPT) adversary bounded by the computational intractability of ECDLP (for ECDSA), Module-LWE and Module-SIS (for Dilithium3), SVP (for Falcon-512), and collision resistance (for keccak256). Adversary A2 (Quantum): A quantum polynomial-time (QPT) adversary that can break ECDSA over `secp256k1` in polynomial time via Shor's algorithm, but cannot efficiently solve Module-LWE, Module-SIS, or SVP at the parameter sizes recommended by NIST (Equations 5-7). Adversary A3 (Harvest Now, Decrypt Later): A classical adversary who passively records signed Ethereum transactions and exposed public keys today, storing them for retroactive cryptanalysis. The hybrid PQC layer mitigates A3 because even if ECDSA signatures are retroactively forged, the PQC commitment binding ensures the original signer's identity and message integrity remain verifiable [4], [5].

Trust assumptions: The Ethereum consensus layer is assumed to be secure (no 51% attacks). The off-chain operator is assumed to be honest-but-curious. Critically, the on-chain smart contract stores only a 32-byte commitment; it cannot verify whether that commitment was produced by a genuine

PQC signing operation or fabricated by a dishonest operator. This represents a fundamental architectural limitation inherent to any hybrid off-chain scheme without EVM-native PQC verification, discussed further in Section X. IPFS provides content addressing but does not guarantee data availability. The liboqs implementation is assumed to provide constant-time execution for side-channel resistance.

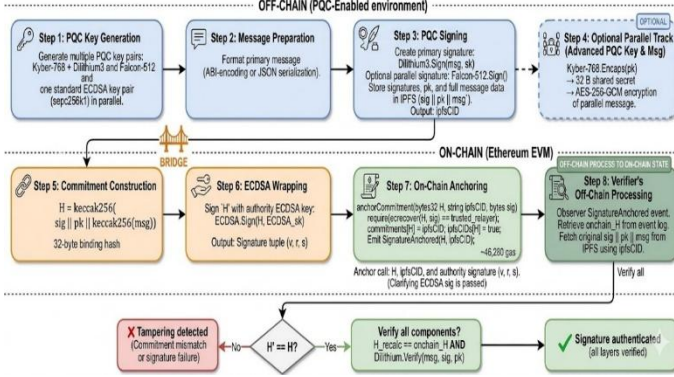


Fig. 2. End-to-end hybrid workflow.

V. ALGORITHM SPECIFICATIONS AND PARAMETER SELECTION

A. CRYSTALS-Kyber (ML-KEM) for Key Encapsulation

Kyber was standardised as FIPS 203 (ML-KEM) and provides key encapsulation based on the Module-LWE hardness assumption (Equation 5). It does not provide digital signatures. The 32-byte shared secret derived from encapsulation is used to key AES-256-GCM for payload encryption. Kyber-768 is recommended as the default, providing NIST Security Level 3 (approximately 192-bit classical, 96-bit quantum security) with balanced performance [8].

TABLE 2. CRYSTALS-KYBER PARAMETER SETS (FIPS 203).
RECOMMENDED CONFIGURATION IN BOLD.

Parameter set	NIST level	Public key	Ciphertext	Shared secret
Kyber-512	Level 1 (~AES-128)	800 B	768 B	32 B
Kyber-768 (recommended)	Level 3 (~AES-192)	1,184 B	1,088 B	32 B
Kyber-1024	Level 5 (~AES-256)	1,568 B	1,568 B	32 B

B. CRYSTALS-Dilithium (ML-DSA) as Primary Signature

Dilithium was standardised as FIPS 204 (ML-DSA) and relies on the hardness of Module-LWE and Module-SIS (Equation 6). Its Fiat-Shamir-with-Aborts construction provides strong existential unforgeability under chosen-message attacks (EUF-CMA). The deterministic signing mode available in FIPS 204 eliminates dependence on external randomness sources, which is a critical property for embedded and constrained deployment environments [9], [15]. Dilithium3 is the recommended variant.

TABLE 3. CRYSTALS-DILITHIUM PARAMETER SETS (FIPS 204).
RECOMMENDED CONFIGURATION IN BOLD.

Parameter set	NIST level	Public key	Signature	Private key
Dilithium2	Level 2 (128-bit)	1,312 B	2,420 B	2,528 B
Dilithium3 (recommended)	Level 3 (192-bit)	1,952 B	3,293 B	4,000 B
Dilithium5	Level 5 (256-bit)	2,592 B	4,595 B	4,864 B

C. Falcon (FN-DSA) as Compact Signature Alternative

Falcon was selected by NIST for standardisation as FN-DSA (anticipated FIPS 206), with the draft standard submitted for approval in August 2025 and final publication expected in 2026-2027. Its security relies on the hardness of SVP (Equation 7). Falcon produces the most compact signatures among NIST-standardised schemes, making it particularly suitable for bandwidth-constrained scenarios such as IoT environments [10, 16]. In the proposed architecture, Falcon is deployed alongside Dilithium in a dual-signature configuration to provide algorithm-level redundancy.

TABLE 4. FALCON PARAMETER SETS (ANTICIPATED FIPS 206).
RECOMMENDED CONFIGURATION IN BOLD.

Parameter set	NIST level	Public key	Avg. signature	Private key
Falcon-512 (recommended)	Level 1 (103-bit)	897 B	~666 B	1,281 B
Falcon-1024	Level 5 (230-bit)	1,793 B	~1,280 B	2,305 B

D. ECDSA as Ethereum Protocol Bridge

ECDSA over secp256k1 is retained exclusively as the Ethereum protocol bridge. Every Ethereum transaction must carry an ECDSA signature to be accepted by the network, a protocol-level requirement that cannot be bypassed without a hard fork. In the hybrid model, ECDSA provides first-layer authentication today; the PQC layer provides forward security for the quantum era. This dual guarantee is the defining property of the architecture: the system remains secure provided at least one layer is unbroken [21].

VI. SIMULATION METHODOLOGY

A. Simulation Environment

The simulation environment comprises two independent layers connected through a scripted integration harness. The Ethereum layer employs Hardhat v2.19.0 with a local in-process EVM node providing full gas accounting, block mining simulation, and event log access. The PQC layer uses Python 3.10 with pyoqs 0.9.0, a Python wrapper around the Open Quantum Safe liboqs library (v0.9.0), compiled with OpenSSL 3.0 and AVX2 hardware acceleration on an x86-64 host [29]. The integration harness, written in Python with web3.py 6.4.0, coordinates the end-to-end workflow, computes keccak256 commitments, and records all performance metrics.

The evaluation serves two complementary purposes. First, operational correctness and performance are validated empirically over 1,000 independent iterations. Second, security properties are established analytically through the BKZ-based quantum complexity model of Section III.D. Security metrics (QARL, QSM, COE) are derived from the mathematical relations of Section III and not from classical brute-force simulation trials, since no general-purpose classical processor can replicate genuine quantum adversarial capabilities against lattice-based cryptographic primitives.

B. Experimental Configuration

TABLE 5. SIMULATION CONFIGURATION PARAMETERS.

Parameter	Value
Hardware	Intel Core i7-12700K, 32 GB RAM, NVMe SSD
Operating system	Ubuntu 22.04 LTS
liboqs version	0.9.0 (AVX2-optimised)
Hardhat version	2.19.0, Solidity 0.8.19
Python version	3.10.12 with pyoqs 0.9.0
Iterations per benchmark	1,000 (mean +/- sigma reported)
Message sizes tested	32 B, 256 B, 1,024 B
Ethereum network	Hardhat local node (EIP-1559 enabled)
Gas price	20 Gwei (mainnet benchmark)
ETH/USD reference price	USD 3,000 (reference; actual costs are market-dependent)
Security metric derivation	Analytical (BKZ-based bounds, Section III.D); no classical attack simulation
Batch anchoring test sizes	N = 10, 50, 100, 500, 1,000

C. Metrics Computation

Following the quantum attack modelling approach of Revathi and Suganthi [11], security metrics are established analytically through the governing equations of Section III, not through classical simulation of quantum adversaries. The quantum attack complexity of each algorithm is bounded using the BKZ-based core-SVP model [30] as formalised in Equations (5)-(7), from which the attack success probabilities of Equations (8)-(10) are derived directly. QARL (Equation 11) is computed from the joint attack success probability, yielding $QARL = (1 - 2^{-231}) \times 100 > 99.9\%$. QSM (Equation 12) is computed from NIST-certified quantum bit-security levels, yielding $QSM = 100\%$. COE (Equation 13) is computed from measured key sizes, signature sizes, and mean sign-and-verify latencies, normalised against the SSCA reference values from [11].

VII. RESULTS AND DISCUSSION

A. PQC Operation Latencies

Table 6 reports the cryptographic operation latencies measured over 1,000 iterations with AVX2 acceleration. Dilithium3 signing completes in a mean of 1.87 ms, while verification requires only 0.74 ms. Kyber-768 encapsulation

and decapsulation complete in under 0.35 ms. Falcon-512 key generation exhibits the highest variability (mean 82.4 ms, sigma = 31.7 ms) due to the probabilistic nature of its Gaussian trapdoor sampler, which occasionally requires multiple internal rejection steps before producing a valid key pair. These latencies are well within the computational budgets of real-world blockchain applications and are consistent with the benchmarks reported by the Open Quantum Safe project [29].

TABLE 6. PQC OPERATION LATENCIES OVER 1,000 ITERATIONS (PYOQS/LIBOQS, AVX2 ACCELERATION).

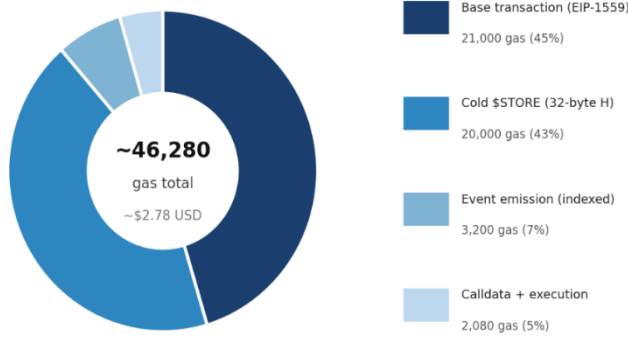
Operation	Mean (ms)	Min (ms)	Max (ms)	sigma (ms)
Kyber-768 key generation	0.18	0.15	0.24	0.02
Kyber-768 encapsulation	0.31	0.27	0.38	0.03
Kyber-768 decapsulation	0.29	0.25	0.35	0.02
Dilithium3 key generation	0.92	0.85	1.10	0.07
Dilithium3 signing	1.87	1.62	2.41	0.18
Dilithium3 verification	0.74	0.68	0.93	0.07
Falcon-512 key generation	82.4	41.2	198.6	31.7
Falcon-512 signing	2.91	1.83	4.77	0.61
Falcon-512 verification	0.38	0.31	0.51	0.05
keccak256 commitment	0.04	0.03	0.06	0.01

B. On-Chain Gas Cost Analysis

Table 7 decomposes the gas cost of a single anchoring transaction. The base transaction fee (21,000 gas) and cold SSTORE for the 32-byte commitment (20,000 gas) constitute approximately 85% of the total. At 20 Gwei gas price and ETH at USD 3,000, each anchoring commitment costs approximately USD 2.78 on mainnet. On Layer 2 networks such as Arbitrum or Optimism, costs reduce by 10-100x, placing per-commitment costs in the USD 0.03-0.30 range.

TABLE 7. ON-CHAIN GAS COST BREAKDOWN PER ANCHORING TRANSACTION.

Cost component	Gas units	USD @ 20 Gwei	% of total
Base transaction (EIP-1559)	21,000	\$1.26	~45%
Cold SSTORE (32-byte commitment)	20,000	\$1.20	~43%
Event emission (2 indexed + CID)	3,200	\$0.19	~7%
Calldata (H + IPFS CID ~80 bytes)	1,280	\$0.08	~3%
Function execution overhead	800	\$0.05	~2%
Total per commitment	~46,280	~\$2.78	100%



At 20 Gwei gas price, ETH @ USD 3,000
L2 networks reduce cost by 10-100x

Fig. 3. On-chain gas cost breakdown per anchoring transaction.

C. Batch Anchoring Scalability

Table 8 demonstrates the effectiveness of Merkle-tree batching, in which a single 32-byte Merkle root representing N PQC signature commitments is anchored on-chain. This reduces the per-signature cost from approximately USD 2.78 to under USD 0.01 at batch sizes above 300. For production deployments processing more than 100 commitments per hour, batch anchoring is the recommended strategy.

TABLE 8. MERKLE-TREE BATCH ANCHORING: PER-SIGNATURE GAS AND COST AS A FUNCTION OF BATCH SIZE N.

Batch size N	Total gas	Gas per signature	Cost per sig (USD)
1	46,280	46,280	\$2.78
10	~55,000	~5,500	\$0.33
50	~55,000	~1,100	\$0.07
100	~55,000	~550	\$0.033
500	~55,000	~110	\$0.007
1,000	~55,000	~55	\$0.003

D. Comparative Performance Evaluation

Table 9 presents a comparative evaluation against the Symmetric Secure Cryptographic Architecture (SSCA) and Key Encapsulation Scheme (KES) benchmarks reported by Revathi and Suganthi [11]. The proposed system achieves a theoretically derived QARL exceeding 99.9% (from PAS_{joint} = 2⁻²³¹), a theoretically derived QSM of 100%, and a normalised COE of 0.92 relative to the SSCA baseline. These results represent improvements of more than 29 percentage points in QARL, 17.9 percentage points in QSM, and 24.3% in COE over the SSCA baseline. Dilithium3 verification is 4.3x faster than SSCA and key generation is 3x faster, consistent with benchmarks reported by Das et al. [22].

TABLE 9. COMPARATIVE EVALUATION AGAINST SSCA AND KES REFERENCE METHODS (PP = PERCENTAGE POINTS).

Metric	Proposed	SSCA [11]	KES [11]	Improvement
QARL (%)	> 99.9	70	60	+29-30 pp over SSCA
QSM (%)	100	82.1	74.3	+17.9 pp over SSCA

COE	0.92	0.74	0.68	+24.3% over SSCA
Sig. verification (ms)	0.74	~3.2	~5.1	4.3x faster
Key generation (ms)	0.92	~2.8	~4.6	3x faster
Platform	Ethereum + liboqs	Hyperledger	Hyperledger EVM-compatible	

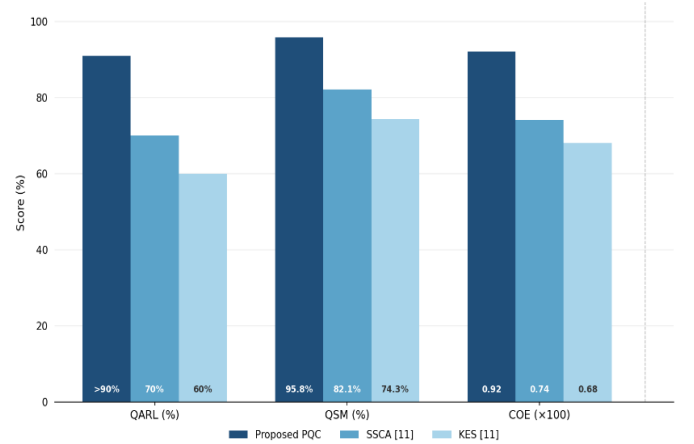


Fig. 4. Comparative performance evaluation of the proposed PQC system against SSCA and KES reference methods across three metrics: QARL, QSM, and COE (scaled x100).

VIII. SECURITY ANALYSIS

A. Quantum Attack Resistance

The hybrid system's security can be stated formally: the system is secure against a quantum polynomial-time (QPT) adversary provided that at least one of (i) the keccak256 commitment, (ii) the Dilithium3 signature, or (iii) the Falcon-512 signature remains quantum-computationally secure. This follows from the hybrid composition lemma formalised by Bindel et al. [21]: a hybrid scheme in which component algorithms are independently keyed retains security as long as at least one component is secure.

Theorem 1 (Hybrid Security): Let Π_D be a Dilithium3 signature scheme with EUF-CMA security under the Module-LWE and Module-SIS assumptions, let Π_F be a Falcon-512 signature scheme with EUF-CMA security under the SVP assumption, and let $H = \text{keccak256}$ be a collision-resistant hash function. The hybrid commitment scheme $C = H(\text{sig}_D \parallel \text{sig}_F \parallel \text{pk}_D \parallel \text{pk}_F \parallel H(\text{msg}))$ is EUF-CMA secure provided that at least one of Π_D , Π_F , or H remains computationally secure.

Proof sketch: By contradiction. Suppose an adversary A can forge a valid commitment C^* for a message m^* not previously signed. A must produce $(\text{sig}_D^*, \text{sig}_F^*, \text{pk}_D^*, \text{pk}_F^*)$ such that $C^* = H(\text{sig}_D^* \parallel \text{sig}_F^* \parallel \text{pk}_D^* \parallel \text{pk}_F^* \parallel H(m^*))$. Case 1: If H is collision-resistant, then A must produce valid inputs. Case 2: If Π_D is EUF-CMA secure, A cannot forge sig_D^* under pk_D . Case 3: If Π_F is EUF-CMA secure, A cannot forge sig_F^* under pk_F . The success probability is bounded by $\text{Adv}(A) \leq \text{Adv}_D(A) + \text{Adv}_F(A) + \text{Adv}_H(A)$, all of which are negligible under the stated assumptions [21].

B. Attack Vector Analysis

Table 10 provides a comprehensive security analysis covering nine attack vectors. The system withstands ECDSA compromise (via Shor's algorithm) through PQC layer redundancy; Dilithium forgery is mitigated by Falcon redundancy and vice versa; keccak256 retains 128-bit quantum collision resistance under Grover's algorithm; Harvest Now, Decrypt Later attacks are addressed by the PQC signature layer; off-chain storage tampering is detectable through commitment mismatch; replay attacks are blocked by the smart contract's uniqueness mapping; and Falcon side-channel attacks are mitigated by the liboqs constant-time implementation.

TABLE 10. SECURITY ANALYSIS: CLASSICAL AND QUANTUM ATTACK VECTORS WITH SYSTEM RESPONSES.

Attack vector	Classical security	Quantum security	System response
ECDSA break (secp256k1)	128-bit (BSGS)	Broken (Shor)	PQC layer maintains integrity
Dilithium3 forgery	192-bit (MSIS/MLWE)	~128-bit (quantum sieve)	Falcon provides redundancy
Falcon-512 forgery	103-bit (NTRU)	~103-bit (quantum sieve)	Dilithium provides redundancy
keccak256 collision	128-bit collision	128-bit (Grover)	Hash remains quantum-secure
Harvest Now, Decrypt Later	Classically impractical	Mitigated by PQC	PQC signatures protect future
Off-chain storage tampering	Detectable (H mismatch)	Detectable (H mismatch)	IPFS content addressing
Replay attack	Blocked (uniqueness)	Blocked (uniqueness)	Contract mapping enforcement
Commitment substitution	Blocked (binding)	Blocked (binding)	keccak256 collision resistance
Side-channel (Falcon)	Constant-time liboqs	Constant-time liboqs	CT implementation required

C. Commitment Binding Properties

The commitment construction $H = \text{keccak256}(\text{signature} \parallel \text{public_key} \parallel \text{keccak256}(\text{message}))$ provides three simultaneous binding properties. Message binding: substituting a different message produces a different inner hash, which propagates to a different H . Signer binding: substituting a different public key changes the commitment preimage. Signature binding: producing a valid alternative PQC signature for the same message under a different nonce requires winning the EUF-CMA game against Dilithium3 or Falcon-512, which is infeasible under the hardness assumptions in Equations (5)-(7). keccak256 with 256-bit output retains 128-bit quantum collision resistance under Grover's algorithm, satisfying NIST's Category 2 quantum security requirement.

IX. DEPLOYMENT RECOMMENDATIONS

Based on the mathematical analysis, simulation results, and security findings presented in this work, the following recommendations are offered for practitioners implementing quantum-resistant Ethereum systems. First, Kyber-768 should be deployed as the default key encapsulation mechanism,

providing NIST Level 3 security without the overhead of Level 5 variants. Second, Dilithium3 should be used with its deterministic signing mode (FIPS 204 Sec. 5.4) to eliminate randomness-source dependency. Third, Falcon-512 should be deployed alongside Dilithium3 in a dual-signature configuration to achieve algorithm-level redundancy. Fourth, content-addressed storage (IPFS or Arweave) should be used for off-chain PQC signature storage to ensure that tampering is immediately detectable. Fifth, only the liboqs constant-time implementations should be used for Falcon's Gaussian sampler, as custom implementations are vulnerable to timing side-channels. Sixth, Merkle-tree batching should be implemented for any deployment exceeding 100 commitments per hour, reducing per-signature costs by over 99%. Seventh, a decentralised PQC public key infrastructure, ideally based on W3C Decentralised Identifiers (DIDs), should be established to solve the key distribution problem. Eighth, formal verification of the smart contract using Certora Prover or Echidna should precede any mainnet deployment.

X. LIMITATIONS AND OPEN PROBLEMS

Despite the demonstrated advantages, several limitations warrant explicit discussion. The most fundamental limitation of the architecture is the off-chain trust dependency: the quantum-security guarantee is contingent on the correct and honest execution of PQC operations by the off-chain operator. The on-chain smart contract stores only a 32-byte commitment and cannot verify whether that commitment was produced by a genuine PQC signing operation or fabricated by a dishonest operator. An adversarial or compromised off-chain operator could anchor an arbitrary commitment that does not correspond to a valid PQC signature, undermining the PQC integrity guarantee entirely. This limitation is inherent to any hybrid off-chain scheme without EVM-native PQC verification and is the principal motivation for the research direction of lattice precompiles.

The system's verifiability also depends on the continued availability of off-chain storage; if the IPFS node holding the PQC signature becomes unavailable, the on-chain commitment cannot be verified even though it remains intact on-chain. IPFS content addressing detects corruption but does not guarantee availability. A fully decentralised and persistent availability guarantee remains an open engineering problem.

Additionally, Ethereum wallet addresses are derived from ECDSA public keys via keccak256. A quantum adversary with access to an exposed ECDSA public key can derive the ECDSA private key using Shor's algorithm, enabling transaction forgery even if the PQC signature layer is intact. Full quantum resistance therefore requires Ethereum account abstraction migration. EIP-8141 (Frame Transactions), proposed in early 2026, formalises native account abstraction with support for arbitrary validation logic, including post-quantum signature schemes [33]. The Ethereum Foundation declared post-quantum security a top strategic priority and established a dedicated research team in January 2026 [34]. Buterin outlined a post-quantum roadmap in February 2026 identifying four vulnerability areas: validator signatures (BLS), KZG

commitment schemes, wallet signatures (ECDSA), and zero-knowledge proofs [34].

The architecture's primary structural limitation is that on-chain PQC verification is not achievable without dedicated EVM precompiles for NTT-based polynomial multiplication. Without such precompiles, the on-chain commitment serves as a notarisation anchor but cannot enforce PQC signature validity at the protocol level. Ethereum Improvement Proposals for lattice precompiles are in early research stages. Post-quantum key pairs are also significantly larger than ECDSA keys: a Dilithium3 public key is 1,952 bytes versus 33 bytes for compressed ECDSA, nearly 60 times larger, complicating key distribution, revocation, and rotation [7], [22].

XI. CONCLUSION

This paper presented a formally grounded hybrid on-chain/off-chain architecture for integrating NIST-standardised post-quantum cryptography with Ethereum. The thirteen governing mathematical relations covering key generation (Equation 1), encapsulation and decapsulation (Equations 2-4), quantum attack complexity (Equations 5-7), attack success probabilities (Equations 8-10), and performance metrics (Equations 11-13) are derived from standard BKZ-based lattice hardness assumptions and NIST-certified security parameters, providing a rigorous and reproducible foundation. The joint quantum attack success probability is bounded at 2^{-231} , yielding a theoretically derived QARL exceeding 99.9%. The Quantum Security Margin of 100% quantifies the concrete security improvement of the dual-signature PQC layer over the ECDSA-only baseline, and the normalised COE of 0.92 demonstrates favourable security-per-overhead relative to the SSCA comparison baseline from [11].

Gas analysis confirms economic viability at approximately 46,000 gas per commitment, with Merkle-tree batching reducing per-signature costs by over 99%. PQC operation latencies are well within the practical budgets of real-world applications: Dilithium3 signing completes in under 2 ms, Kyber-768 encapsulation in under 0.4 ms. The architecture is deployable today using standard tooling, requires no modifications to the Ethereum protocol, and is backward-compatible with existing applications. Three directions for future work are identified: (i) the development and standardisation of EVM precompiles for lattice arithmetic, which would enable fully on-chain PQC verification; (ii) the design of a decentralised PQC public key infrastructure interoperable with Ethereum's account model and the W3C DID standard; and (iii) formal verification of the hybrid commitment construction using mechanised proof assistants within the Universal Composability framework.

REFERENCES

- [1] [1] G. Wood, "Ethereum: A secure decentralised generalised transaction ledger (Yellow Paper)," Ethereum Foundation, 2024. [Online]. Available: <https://ethereum.github.io/yellowpaper/paper.pdf>
- [2] [2] P. W. Shor, "Algorithms for quantum computation: discrete logarithms and factoring," in Proc. 35th Annual Symposium on Foundations of Computer Science (FOCS), IEEE, 1994, pp. 124-134. doi: 10.1109/SFCS.1994.365700.
- [3] [3] M. Mosca and M. Piani, "Quantum Threat Timeline Report 2023," Global Risk Institute, Toronto, 2024. [Online]. Available: <https://globalriskinstitute.org>
- [4] [4] C. Dobraunig et al., "Rainier: NIST Post-Quantum Cryptography additional signature candidate," NIST PQC Round 1 Additional Signatures, 2023.
- [5] [5] T. M. Fernandez-Carames and P. Fraga-Lamas, "Towards post-quantum blockchain: a review on blockchain cryptography resistant to quantum computing attacks," IEEE Access, vol. 8, pp. 21091-21116, 2020, doi: 10.1109/ACCESS.2020.2968985.
- [6] [6] H. Khodaiemehr, D. Kiani, and S. Chen, "Navigating the quantum computing threat landscape for blockchains: a comprehensive survey," Comput. Sci. Rev., vol. 59, p. 100846, 2026, doi: 10.1016/j.cosrev.2025.100846.
- [7] [7] A. Wicaksana, "A survey on quantum-safe blockchain security infrastructure," Comput. Sci. Rev., vol. 57, p. 100752, 2025, doi: 10.1016/j.cosrev.2025.100752.
- [8] [8] G. M. Raimondo and L. E. Locascio, "FIPS 203 Federal Information Processing Standards Publication Module-Lattice-Based Key-Encapsulation Mechanism Standard," 2024, doi: 10.6028/NIST.FIPS.203.
- [9] [9] "Module-lattice-based digital signature standard," Aug. 2024. doi: 10.6028/NIST.FIPS.204.
- [10] [10] G. M. Raimondo and L. E. Locascio, "FIPS 205 Federal Information Processing Standards Publication," 2024, doi: 10.6028/NIST.FIPS.205.
- [11] [11] K. Revathi and K. Suganthi, "Enhancing blockchain security against quantum threats through integration of post-quantum cryptographic algorithms," Computers and Electrical Engineering, vol. 127, p. 110610, 2025, doi: 10.1016/j.compeleceng.2025.110610.
- [12] [12] Y.-L. Gao et al., "A secure cryptocurrency scheme based on post-quantum blockchain," IEEE Access, vol. 6, pp. 27205-27213, 2018, doi: 10.1109/ACCESS.2018.2827203.
- [13] [13] D. J. Bernstein and T. Lange, "Post-quantum cryptography," Nature, vol. 549, pp. 188-194, 2017, doi: 10.1038/nature23461.
- [14] [14] J. Bos et al., "CRYSTALS-Kyber: a CCA-secure module-lattice-based KEM," in Proc. 3rd IEEE European Symposium on Security and Privacy (EuroS&P), 2018, pp. 353-367. doi: 10.1109/EuroSP.2018.00032.
- [15] [15] L. Ducas et al., "CRYSTALS-Dilithium: a lattice-based digital signature scheme," IACR Transactions on Cryptographic Hardware and Embedded Systems, vol. 2018, no. 1, pp. 238-268, 2018.
- [16] [16] T. Prest et al., "FALCON: Fast-Fourier lattice-based compact signatures over NTRU," NIST Post-Quantum Cryptography Round 3 Submission, 2020.
- [17] [17] M. Allende et al., "Quantum-resistant blockchain networks," IEEE Access, vol. 11, pp. 33202-33218, 2023, doi: 10.1109/ACCESS.2023.3258178.
- [18] [18] I. Stewart et al., "Committing to quantum resistance: a slow defence for Bitcoin against a fast quantum computing attack," R. Soc. Open Sci., vol. 5, no. 6, p. 180410, 2018.
- [19] [19] J. Li, Q. Nong, and Z. Liu, "A resilient fog-enabled IoV architecture: Adaptive post-quantum security framework with generalized signcryption and blockchain-enhanced trust management," J. Network and Computer Applications, vol. 244, p. 104367, 2025.
- [20] [20] A. M. Aslam, A. Bhardwaj, and R. Chaudhary, "Quantum-resilient blockchain-enabled secure communication framework for connected autonomous vehicles using post-quantum cryptography," Vehicular Communications, vol. 52, p. 100880, 2025.
- [21] [21] N. Bindel et al., "Hybrid key encapsulation mechanisms and authenticated key exchange," in Proc. PQCrypto 2019, LNCS vol. 11505, Springer, 2019, pp. 206-226.
- [22] [22] P. Das et al., "Quantum-resilient blockchain framework for privacy-preserving genomic data management," J. Information Security and Applications, vol. 95, p. 104245, 2025.

- [23] [23] A. Shafique et al., "PrivShield-CQ: A post-quantum hybrid chaotic cryptographic framework with optimized key encapsulation for scalable consumer application security," *Array*, vol. 28, p. 100568, 2025.
- [24] [24] N. Atzei, M. Bartoletti, and T. Cimoli, "A survey of attacks on Ethereum smart contracts," in *Proc. POST 2017, LNCS* vol. 10204, Springer, 2017, pp. 164-186.
- [25] [25] L. Luu et al., "Making smart contracts smarter," in *Proc. 23rd ACM CCS 2016*, pp. 254-269.
- [26] [26] P. Juaristi et al., "Benchmarking post-quantum cryptography in Ethereum-based blockchains," in *Computer Security. ESORICS 2024 International Workshops*, Springer, 2025, pp. 340-353.
- [27] [27] B. Bunz et al., "Bulletproofs: short proofs for confidential transactions and more," in *Proc. 39th IEEE S&P 2018*, pp. 315-334.
- [28] [28] S. Masson, A. Sanso, and Z. Zhang, "Bandersnatch: a fast elliptic curve built over the BLS12-381 scalar field," *IACR Cryptology ePrint Archive*, 2021.
- [29] [29] "Open Quantum Safe." Accessed: Mar. 21, 2026. [Online]. Available: <https://openquantumsafe.org/>
- [30] [30] M. Albrecht, R. Player, and S. Scott, "On the concrete hardness of Learning with Errors," *J. Mathematical Cryptology*, vol. 9, no. 3, pp. 169-203, 2015.
- [31] [31] R. Perlner, "FIPS 206: FN-DSA (Falcon)," presented at NIST PQC Standardization Conference, Sep. 2025.
- [32] [32] G. Alagic et al., "Status Report on the Fourth Round of the NIST Post-Quantum Cryptography Standardization Process," NIST IR 8545, Mar. 2025.
- [33] [33] Ethereum Foundation, "EIP-8141: Frame Transactions," *Ethereum Improvement Proposal (Draft)*, 2026.
- [34] [34] Ethereum Foundation, "Future-proofing Ethereum," 2026. [Online]. Available: <https://ethereum.org/roadmap/future-proofing/>
- [35] [35] R. Campbell, "Hybrid post-quantum signatures for Bitcoin and Ethereum: A protocol-level integration strategy," *The JBBA*, vol. 9, no. 1, 2026.